4.6.1 Fleming's left-hand rule

Hold the left thumb, first finger and second finger mutually perpendicular. The first finger indicates the external magnetic field, the second conventional current, and the thumb magnetic force. For particles, use positive-charge motion as current direction and reverse the result for negative charge. This force rule differs from the right-hand rule for the field produced by a current.

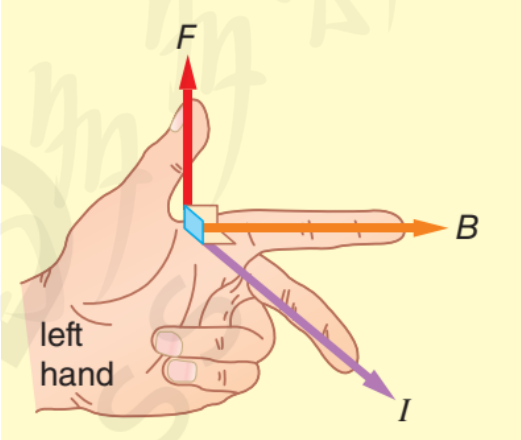
Force on a current-carrying wire

A straight wire in a uniform magnetic field experiences force depending on flux density, current, length inside the field and the angle between current and field. Force is greatest when perpendicular and zero when parallel. The left-hand rule gives direction. Flux density is measured in tesla. Exclude wire length outside the field.

$$1\ \mathrm{T} = 1\ \mathrm{N\,A^{-1}\,m^{-1}}$$
$$F = BIl\sin\theta$$

Tesla is magnetic flux density's SI unit; from F = BIl with the wire perpendicular to the field.

F: force magnitude (N); B: magnetic flux density (T); I: conventional current (A); l: wire length in field (m); θ: angle between current direction and B. Uniform field and straight wire.

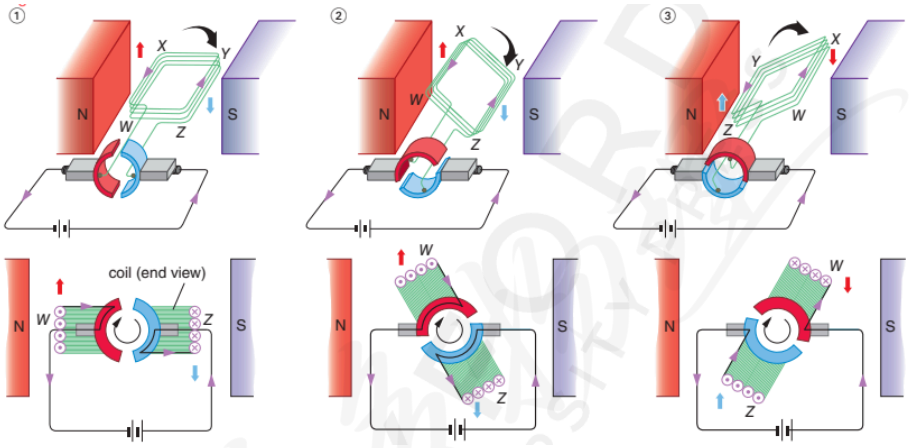


First finger B, second finger I and thumb F are mutually perpendicular. I means conventional current; a negative moving charge reverses the force obtained from its velocity.

4.6.2 The d.c. motor

Opposite sides of a current-carrying coil experience opposite magnetic forces, creating a turning moment and converting electrical energy to mechanical energy. A split-ring commutator reverses coil current every half-turn, so the moment continues to drive rotation in the same sense after the coil passes the relevant position.

In the simple uniform-field model, the active sides remain perpendicular to the field. With B, I and l fixed, their force BIl stays constant, but the lever arm and moment change with coil angle. At zero lever arm, the rotating coil passes the zero-moment position through inertia, then experiences a driving moment again.



The commutator reverses coil current every half-turn to maintain the driving sense of rotation.

4.6.3 Forces between parallel currents

Each wire produces a magnetic field acting on the other. Parallel currents in the same direction attract. Opposite currents repel. Forces are equal in magnitude, and force per unit length for long wires is proportional to the current product and inversely proportional to separation.

$$\frac{F}{l} = \frac{\mu_0 I_P I_Q}{2\pi r}$$

Parallel long wires: force per length (N m^{-1}); current magnitudes I_P, I_Q (A); separation r (m). Same current direction: attraction; opposite: repulsion.

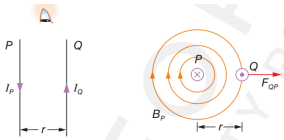


Fig 6.3a *P and Q carrying currents in opposite directions.*

Fig 6.3b *The magnetic force acting on Q due to P.*

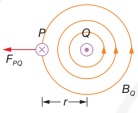


Fig 6.3c *The magnetic force acting on P due to Q.*

Conversely, if I_P and I_Q are in the same direction, P and Q will attract each other (Fig 6.3d–6.3f).

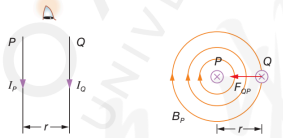


Fig 6.3d *P and Q carrying currents in the same direction.*

Fig 6.3e *The magnetic force acting on Q due to P.*

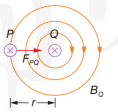


Fig 6.3f *The magnetic force acting on P due to Q.*

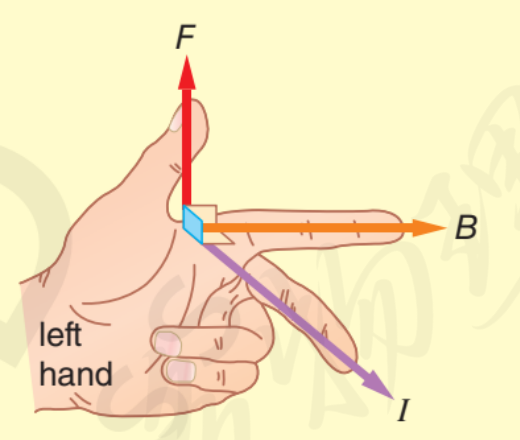
Use the field of one wire to find the force on the other. Opposite currents repel (top); currents in the same direction attract (bottom).

4.6.4 Magnetic force on a moving charge

Magnetic force is perpendicular to velocity and field and depends on the velocity component perpendicular to the field. Motion parallel to the field gives zero magnetic force. Opposite charge signs give opposite force directions. Magnetic force alone does no work, changing direction rather than speed. Perpendicular entry into a uniform field can produce uniform circular motion.

$$F = |Q|vB\sin\theta$$

F: magnetic-force magnitude (N); Q: charge (C); v: speed (m s^{-1}); B: T; θ: angle between velocity and field. Direction reverses for negative charge; the magnetic force does no work.



First finger B, second finger I and thumb F are mutually perpendicular. I means conventional current; a negative moving charge reverses the force obtained from its velocity.